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A

Connector, input surge protection, chassis

File: connector.kicad_sch

eFuse, buck 5V, LDO 3.3V

File: power.kicad_sch

MCU STM32L431, SWD, status LED

File: mcu.kicad_sch

B

Capacitance (PCAP04) and Pt1000 (ADS1220)

File: measurement.kicad_sch

Analog outputs x2 (V/I selectable) with miswiring protection

File: analog_out.kicad_sch

RS-485 (+/-70V fault protected)

File: rs485.kicad_sch

C

D

E

F

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DOTECH HMT500 Oil Moisture Transmitter – Schematic v0.1

Outputs: RS-485 Modbus RTU + 2x analog (V/I selectable). Supply 12-30 V DC.

Protection: any pin pair +/-30 V continuous miswiring, surge +/-1 kV (design target).

Connections are made with global net labels. Placement/wiring cleanup is next step in KiCad.

Parts with field VERIFY=YES (TPS2660, LMR36006, PCAP04, DAC8760, TPS26611) have PLACEHOLDER pin numbers.

TBD footprints: M Connect 8P, CMC, GDT, feedthrough, inductor, placeholder-pinout ICs.

Symbols with VERIFY=YES have placeholder pin numbers
Generated by hardware/kicad/gen_hmt500.py – see docs/hw/circuit-design.md
Oil moisture transmitter HMT500
DOTECH Co., Ltd.
Sheet: /
File: HMT500.kicad_sch

Title: HMT500 – Top

Size: A3Date: 2026-09-26

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4Id: 1/7

Rev: 0.1











